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ECONOMIC FREEDOM AND INCOME LEVELS ACROSS U.S. STATES: A SPATIAL PANEL DATA ANALYSIS

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There is a large literature estimating the effect of economic freedom on economic growth or income levels. Most studies examine the relationship between economic freedom and growth or income levels for countries, while a few examine the relationship for U.S. states. Absent in the state-level literature is consideration of the presence of spatial spillovers affecting the freedom-income relationship. Neglecting to account for spatial autocorrelation can bias estimation results and therefore inferences drawn. We find evidence of a spatial pattern in real per-capita gross state product (GSP) that affects nonspatial estimates of the freedom-income relationship. Taking into account the direct and indirect effects of economic freedom on real per-capita GSP, we find a 10% increase in economic freedom is associated with a 5% increase in real per-capita GSP. (*JEL* E02, O47, R11)

I. INTRODUCTION

The role that market-oriented institutions play in determining long-run economic prosperity has recently come to the forefront of the economics profession. In recent years the development of cross-country measures of institutions, such as the Economic Freedom of the World (EFW) index (Gwartney, Lawson, and Hall 2015), has allowed scholars to analyze the relationship between institutions and growth using panel data methods over more recent and shorter time frames. Important early works in this literature include Easton and Walker (1997) and Dawson (1998), who largely find that economic freedom is positively associated with income levels or economic growth. The creation of the Economic Freedom of North America (EFNA) index (Stansel, Torra, and McMahon 2014) has allowed scholars to test the economic freedom-growth/income relationship within the United States and these studies find similar results (Compton, Giedeman, and Hoover 2011; Wiseman and Young 2013).

A critical survey of the economic freedom-growth/incomes relationship by De Haan et al. (2006, 157) concluded that "there are strong indications that liberalization, i.e., an increase in the EF index, stimulates economic growth." In terms of economic freedom and income levels, papers by Cebula, Clark, and Mixon Jr (2013), Cebula and Clark (2014), Bennett (2016), and Foley and Clark (2016) find the same positive relationship between economic freedom and income levels controlling for a diverse set of factors and over varying time periods for countries and U.S. states. Given the consistency of

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ABBREVIATIONS

BEA: Bureau of Economic Analysis
EFNA: Economic Freedom of North America
EFW: Economic Freedom of the World
GMM: Generalized Method of Moments
GSP: Gross State Product
LM: Lagrange Multiplier
LR: Likelihood Ratio
MSA: Metropolitan Statistical Area
NAICS: North American Industry Classification System
OLS: Ordinary Least Squares
SAR: Spatial Autoregressive
SDM: Spatial Durbin Model
SEM: Spatial Error Model
SIC: Standard Industrial Classification

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this subsequent literature, it is not surprising that summaries of the empirical literature citing the EFW by Hall and Lawson (2014) and the subnational level by Hall, Stansel, and Tarabar (2015) found only one paper suggesting that the EFW index is not robustly related to economic growth or income levels. The relationship between economic freedom and economic prosperity appears to be as strong as any empirical relationship given the wide variety of empirical specifications and approaches that have been used to test this relationship.

To this point, however, no scholars have accounted for spatial dependence in variables across countries or states when analyzing the economic freedom-growth relationship in a panel data setting. This is not surprising given that much of the economics literature has focused on causality (Murphy 2016). From a policy perspective, however, economists should also care about "oomph" (Ziliak and McCloskey 2016). There are strong theoretical reasons based on Tiebout (1956) and yardstick competition (Brueckner and Saavedra 2001) to think that economic policies are spatially related. Recent empirical work by Leeson and Dean (2009) shows that democracy and economic freedom spread geographically across countries, that is, that changes in a country's economic or political institutions "spill over" to nearby countries. Similarly, a number of papers using the EFNA as an explanatory variable find evidence of spatial dependence (Ashby 2007; Mulholland and Hernández-Julián 2013). Appropriately accounting for spatial dependence where it exists is important because the failure to do so can lead to incorrect conclusions, which is especially important for policy-relevant research (LeSage and Dominguez 2012).

In this paper we make two contributions. The first is to the literature on economic freedom and income levels. Ours is the first state-level paper to deal with spatial dependence in the economic freedom-income relationship by using spatial econometric techniques.[1] We find evidence of spatial autocorrelation in real per-capita gross state product (GSP) across U.S. states. Failure to account for this spatial dependence means that nonspatial papers are providing incorrect estimates of the true marginal effects of institutions on income levels. This has important implications for policies, like tax reductions, that may represent a trade-off between current spending and the size of the future economic pie. Second, ours is the first paper in the burgeoning state economic freedom literature (Apergis, Dincer, and Payne 2014; Barnatchez and Lester 2017; Heller and Stephenson 2014; Hoover, Compton, and Giedeman 2015) to employ spatial panel data econometric estimation techniques. In doing so we highlight how spatial panel models allow for the estimation of indirect or "spillover" effects of changes in explanatory variables. We find that after accounting for both the direct and indirect effects of state-level economic freedom on real per-capita GSP, a 10% increase in economic freedom increases real per-capita GSP by 5%.

Our paper proceeds as follows. While we focus on income levels and not changes in growth, we follow Compton, Giedeman, and Hoover (2011) in our choice of explanatory variables, including the use of the EFNA index as our measure of economic freedom.[2] We generally follow Compton, Giedeman, and Hoover (2011) in our choice of explanatory variables as it is the best cited and most frequently referenced paper in the recent literature on institutions and growth/income levels (Bjørnskov 2017; Hoover, Compton, and Giedeman 2015). After briefly describing the data, we describe spatial panel models. The following section presents our empirical results, including results where we break up economic freedom into its three component areas: Size of Government, Takings and Discriminatory Taxation, and Regulation. We conclude with some thoughts on our results and the implications for the economic freedom-economic growth/incomes literature specifically and the economic freedom literature more generally.

II. DATA AND ECONOMETRIC APPROACH

In this paper, we focus on the effect of state-level economic freedom on income levels, measured by the log of real per-capita GSP. We focus on state-level economic freedom for

1. Bologna, Young, and Lacombe (2016) find spatial dependence in the economic freedom-income levels relationship in a cross-section of U.S. MSAs. Wiseman (2017) finds spatial dependence in the growth of incomes of the top 10% and bottom 90% of the income distribution.

2. Compton, Giedeman, and Hoover (2011) estimate both OLS and System GMM dynamic panel models, generally getting more significant results on the economic freedom variable in the OLS models. The overall economic freedom scores show a significantly positive relationship with economic growth in their paper. The spatial nature of our approach does not allow us to use System GMM.

two reasons. The first is pragmatic based on data limitations. While there are a number of papers that focus on metropolitan statistical area (MSA)-level growth (Bologna, Young, and Lacombe 2016; Glaeser, Kerr, and Kerr 2015; Samila and Sorenson 2011), there is only 1 year of data on MSA-level economic freedom (Stansel 2013). There exists no county-level measure of economic freedom. The second reason for focusing on state-level economic freedom is that in the U.S. context, the real variation in economic freedom across areas occurs as a result of state policy. Cities and counties are creatures of state constitutions even in home rule states, and therefore have limits on their ability to vary in terms of economic freedom.[3]

The model used in our estimations mirrors that found in Compton, Giedeman, and Hoover (2011), though ours includes state-level data from 1981 through 2012 and, because of the spatial techniques discussed below, omits data for Alaska and Hawaii. The dependent variable is the log of real per-capita GSP obtained from the Bureau of Economic Analysis (BEA).[4] Independent variables obtained from the Census Bureau include education (percentage of the population 25 and older that graduated from college), percent black, and percent Hispanic. The percentage of the population living in metropolitan areas was obtained from the BEA Local Area CA1 Personal Income Summary table. Real per-capita gross private domestic investment by state is the real (in 2009 dollars) national gross private domestic investment multiplied by the ratio of the state-to-national (nominal) personal income, divided by state population. The ratio of incomes at the 90th percentile to the incomes by the 10th percentile was obtained from Voorheis (2014). The freedom index variable is the overall score at the state and local level (Table 3.4c in Stansel, Torra, and McMahon 2014). Table 1 provides summary statistics for the data used.

3. To give but one example, while 45 states utilize the income tax, only 14 states and the District of Columbia let substate jurisdictions levy income taxes Moreno (2017).

4. Since GSP is reported using Standard Industrial Classification (SIC) through 1997, and using North American Industry Classification System (NAICS) from 1997 and later, we adjusted the earlier SIC values by multiplying them by the ratio of the 1997 values (1997 NAICS/1997 SIC). This follows the approach employed by Compton, Giedeman, and Hoover (2011). Values are in 2009 dollars, adjusted using the gross domestic product chain-type price index.

TABLE 1
Summary Statistics

Variable	Mean	Standard Deviation	Min	Max
Real per-capita GSP	37,507	9,755	18,199	79,132
Education	0.22	0.05	0.10	0.39
Investment	6,114	1,878	2,504	12,984
Economic freedom	6.72	0.69	4.04	8.55
Size of government	7.31	0.96	4.26	9.59
Takings and discriminatory taxation	6.53	0.76	3.45	8.65
Regulation	6.31	1.07	2.16	8.66
Black	0.10	0.09	0.00	0.37
Hispanic	0.07	0.08	0.00	0.47
Metro	0.74	0.19	0.29	1.00
90/10 income inequality	7.48	4.22	4.22	13.76

A family of related spatial econometric models can be represented by the following[5]:

$$(1)\quad y_{it} = \delta \sum_{j=1}^{N} w_{ij} y_{jt} + x_{it}\beta + \sum_{j=1}^{N} w_{ij} x_{ijt}\gamma + \mu_i + \lambda_t + u_{it}$$

$$u_{it} = \rho \sum_{j=1}^{N} w_{ij} u_{it} + \varepsilon_{it}$$

where i is an index for the cross-sectional dimension (i.e., the spatial units, or states in the United States), with $i = 1, \ldots N$, and t is an index for the time dimension (i.e., the time periods), with $t = 1, \ldots T$. The dependent variable y_{it} is an observation on the dependent variable at i and t, x_{it} is a $(1 \times K)$ row vector of observations on the explanatory variables, and β is a matching $(K \times 1)$ vector of fixed but unknown parameters. The terms μ_i and λ_t represent space- and time-period fixed effects, respectively.

The additional terms in the above equation are what make the panel model a spatial econometric one. In particular, the model may contain a spatially lagged dependent variable or a spatial autoregressive (SAR) process in the error term. In addition, there may be spatially weighted explanatory variables in the model. An important aspect of any spatial econometric model is the spatial arrangement of the units in the sample. In practice, this is accomplished by specifying a spatial weight matrix, W. The individual elements in the spatial weight matrix, w_{ij}, would equal "1" if observations i and j were "neighbors" (based

5. Throughout this section we utilize the notation in Elhorst (2014b).

on some metric) and "0" otherwise. Normally, a row stochastic weight matrix is used in a regression modeling context, which means that the rows of the spatial weight matrix sum to unity. This transformation of the spatial weight matrix provides for an intuitive explanation for the *Wy* and *Wu* terms. The *Wy* term can be thought of as a weighted average of the surrounding observations on the dependent variable, and *Wu* can be thought of as a weighted average of the surrounding error terms. Depending on the regression modeling context, both δ and ρ measure the extent of the spatial autocorrelation, in the dependent variable and the error term, respectively.

Given Equation (1), special cases can be obtained by restricting parameters. For example, setting both $\rho = 0$ and $\gamma = 0$, we obtain a model that exhibits spatial dependence only in the dependent variable. This model is the SAR model which can be expressed mathematically as follows:

$$y_{it} = \delta \sum_{j=1}^{N} w_{ij} y_{jt} + x_{it}\beta + \mu_i + \lambda_t + \varepsilon_{it}. \tag{2}$$

The spatial error model (SEM) arises when the restrictions $\delta = 0$ and $\gamma = 0$ are in effect, resulting in spatial dependence in the error term alone. The SEM model can be expressed mathematically as follows:

$$y_{it} = \sum_{j=1}^{N} x_{it}\beta + \mu_i + \lambda_t + u_{it}$$

$$u_{it} = \rho \sum_{j=1}^{N} w_{ij} u_{it} + \varepsilon_{it}. \tag{3}$$

Placing the restriction $\rho = 0$ results in the spatial Durbin model (SDM). The SDM allows for a spatially lagged dependent variable as well as spatially lagged independent variables. The spatially lagged independent variables are explanatory variables that are premultiplied by the spatial weight matrix and represent a weighted average of the surrounding values of the explanatory variables. In other words, the spatially weighted explanatory variables capture any spillover effects that may be present. The SDM model can be expressed as follows:

$$y_{it} = \delta \sum_{j=1}^{N} w_{ij} y_{jt} + x_{it}\beta + \sum_{j=1}^{N} w_{ij} x_{ijt}\gamma + \mu_i + \lambda_t + \varepsilon_{it}. \tag{4}$$

The SAR model is used when one believes that there may be possible spatial autocorrelation in the dependent variable. It is important to note that the inclusion of the *Wy* term on the right-hand side of the above equation introduces simultaneity bias and therefore the use of ordinary least squares (OLS) as an estimation strategy will produce biased and inconsistent parameter estimates (LeSage and Pace 2009). Therefore maximum likelihood estimation is used to estimate the parameters in the SAR model.[6]

The SEM is utilized when one believes that there may be variables that are omitted from the model that are spatially correlated but that are uncorrelated with the included regressors. The conditions under which this spatial residual autocorrelation arises are nicely illustrated in a housing context by Dubin (1998, 304): "Housing prices are a prime example: clearly the location of the house will have an effect on its selling price. If the location of the house influences its price, then the possibility arises that nearby houses will be affected by the same location factors. Any error in measuring these factors will cause their error terms to be correlated." In the SEM, the OLS estimator is unbiased, but inefficient. SEM can also be efficiently estimated via maximum likelihood.

LeSage and Pace (2009) point out that SDM should be used when one believes that there are omitted variables in the model that are spatially correlated and these spatially correlated omitted variables are correlated with an included explanatory variable in the model. If these two conditions hold, the SDM is the most appropriate model. As indicated by Equation (1) all three of these models may include space- and time-fixed effects; in our case we have state- and year-fixed effects. To determine whether such fixed effects are jointly significant, standard likelihood ratio (LR) tests can be performed. Further details regarding the estimation and use of spatial panel data models are contained in Elhorst (2014a).

A limitation of our empirical approach is that we cannot address the possibility of reverse causality. That is, it could be income levels leading to changes in the level of economic freedom, not economic freedom being associated with higher income levels (Islam 1996). Unfortunately, our spatial approach precludes traditional approaches to addressing this issue such as

6. Details regarding maximum likelihood estimation of spatial econometric models are contained in LeSage and Pace (2009).

System generalized method of moments (GMM) and instrumental variable approaches.

III. EMPIRICAL RESULTS

Many panel data models that examine the effect of economic freedom on growth (e.g., Compton, Giedeman, and Hoover 2011) include both state- and year-fixed effects. To determine if both types of these fixed effects should be included in our empirical specification, we conduct two LR tests: one for the inclusion of state fixed effects and one for the inclusion of time (or in our case, year) fixed effects. The null hypothesis for the state fixed effects is H_o: μ_1, μ_2, $\ldots$, $\mu_n = 0$ and the results indicate that this null hypothesis should be rejected (LR: 2160.57, 48 df, p value .0000). The null hypothesis for the year fixed effects is H_o: $\lambda_1, \lambda_2, \ldots, \lambda_t = 0$ and the results indicate that this hypothesis should also be rejected (LR: 513.55, 25 df, p value .0000). Therefore, both state- and year-fixed effects are included in our models.

Another factor that must be considered is the type of spatial econometric model to utilize, that is, should one use the SAR, SEM, or SDM model? Elhorst (2010) has proposed a testing procedure that (1) uses the SDM as the point of departure and (2) builds on the earlier work of Florax, Folmer, and Rey (2003). The first step in the procedure is to use the standard Lagrange multiplier (LM) tests to determine if the SAR or SEM model is appropriate. The LM lag test is a test to determine if there is any omitted spatial autocorrelation in the dependent variable, while the LM error test is designed to detect any residual spatial error correlation. Both of these tests are one-way tests in that they only take into account the specific type of spatial dependence being tested. Additionally, each of these test statistics is distributed under the null hypothesis of no spatial autocorrelation χ^2 with one degree of freedom. The robust varieties of these tests (i.e., the LM lag robust and LM error robust tests) are designed to take into account the other type of spatial dependence: that is, the LM lag robust test takes into account the possible presence of spatial dependence in the error term while the LM error robust test takes into account the possibility of spatial dependence in the dependent variable. A rubric for determining the proper spatial econometric model is contained in Florax, Folmer, and Rey (2003).

TABLE 2
LM Test Results

LM Test	LM Test Value	p Value
LM lag	3.99	0.045
LM error	11.97	0.000
LM lag robust	0.21	0.641
LM error robust	8.19	0.004

Table 2 contains results from the various LM tests and shows that the null hypothesis of no spatial correlation in the dependent variable or the error term can be safely rejected. Elhorst (2010), who has developed a more complete testing procedure, at this point recommends that the SDM be estimated and that the two following hypotheses be tested via a LR test:

$$H_o : \theta = 0 \tag{5}$$

$$H_o : \theta + \delta\beta = 0. \tag{6}$$

The first null hypothesis is a test to determine if the SDM can be reduced to the SAR model and the second hypothesis tests whether the SDM can be reduced to the SEM model. Both the first null hypothesis (LR 49.57, p value .0000) and the second null hypothesis (LR 38.66, p value .0000) can be rejected at the 1% level of significance. In this instance, Elhorst (2010) recommends that the SDM be estimated and the null hypothesis H_o: $\delta = 0$ be tested. The results of this test indicate that the null hypothesis is rejected at the 1% level of significance ($\delta = -0.27$, t-stat -5.10, p value .0000) and we conclude from our testing procedure that the spatial Durbin panel model is the most appropriate model to use in the empirical analysis.[7]

The results of the SDM model are contained in Table 3. The first noteworthy result is that the spatial autocorrelation parameter δ is negative, with a value of -0.094 and statistically significant at the 1% level. This indicates a modest, although highly statistically significant, amount of spatial autocorrelation in our dependent variable. These results indicate that a simple panel data model that did not take into account the spatial dependence would lead to incorrect inferences.

Regression models that contain a spatially lagged y variable (i.e., a ρWy term) have coefficients estimates that are not directly interpretable.

7. According to LeSage and Pace (2009) the SDM model is the only model that produces unbiased coefficient estimates under three other spatial econometric data generating processes.

TABLE 3
Spatial Durbin Panel Model Results

	Coefficient	*t*-stat	*p* Value	
Direct				
Education	0.157	3.89	.00	***
Investment	1.210	42.54	.00	***
Economic freedom	0.031	1.12	.26	
Black	0.029	3.50	.00	***
Hispanic	0.017	3.08	.00	***
Metro	−0.247	−4.67	.00	***
90/10 income inequality	0.012	1.41	.16	
Indirect				
Education	−0.157	−1.92	.05	*
Investment	−0.087	−1.32	.19	
Economic freedom	0.473	5.58	.00	***
Black	−0.062	−3.76	.00	***
Hispanic	0.012	0.88	.37	
Metro	0.446	5.08	.00	***
90/10 income inequality	0.060	2.73	.00	***
Total				
Education	−0.000	−0.00	.99	
Investment	1.122	15.58	.00	***
Economic freedom	0.504	5.22	.00	***
Black	−0.033	−1.93	.05	*
Hispanic	0.030	1.93	.05	*
Metro	0.199	2.33	.02	**
90/10 income inequality	0.073	2.94	.00	**
Spatial autocorrelation δ	0.094	3.92	.00	***
R^2	.98			
Observations	1,536			

Notes: Dependent variable is real per-capita GSP. All variables are in log form.
* Significant at the 10% level, ** significant at the 5% level, *** significant at the 1% level.

We can formally represent this by writing the SDM in reduced form as follows

$$y = \delta W y + X\beta + WX\gamma + \varepsilon \tag{7}$$

$$y = \left(I_n - \delta W\right)^{-1}(X\beta + WX\gamma) + \left(I_n - \delta W\right)^{-1}\varepsilon \tag{8}$$

$$S(W) = \partial y/\partial x_r = \left(I_n - \delta W\right)^{-1}(\beta + W\gamma). \tag{9}$$

The multiplication of the $S(W)$ matrix by our coefficient estimates results in an $N \times N$ matrix of effects estimates, whereby the diagonal elements represent the direct plus feedback effects and the off-diagonal elements represent the indirect or spillover effects.[8] Mathematically, we can express each of these partial derivative effects as follows:

$$\text{Direct Effect}: \frac{\partial E\left(y_i\right)}{\partial x_i} = S(W)_{ii} \tag{10}$$

$$\text{Indirect Effect}: \frac{\partial E\left(y_i\right)}{\partial x_j} = S(W)_{ij}. \tag{11}$$

The direct plus feedback effect is the marginal effect of a change in an independent variable at location i and how it affects the dependent variable at location i, while the indirect effect is how a change in an independent variable at location j affects the dependent variable at location i, where $i \neq j$.

Given the potentially large numbers of observations that can be analyzed, LeSage and Pace (2009) recommend several scalar summary measures of the direct and indirect effects.[9] The average direct effect is taken to be the average of the main diagonal of the $S(W)$ matrix, while the average total effects are the average of the row sums of the $S(W)$ matrix. The average indirect effects are the difference between the average total effects and the average direct effects.

The effects estimates calculated from a model that contains a spatially lagged y variable provide

8. Note that each explanatory variable will have an associated $N \times N$ matrix of effects estimates.

9. For example, in a county-level data setting, the $S(W)$ matrix would consist of a 3,061 × 3,061 matrix of effects.

a much richer set of results that can be analyzed compared to a nonspatial panel data model, which by assumption does not include any spillover effects as a consequence of the independence of observations assumption. Another advantage of using the SDM model is that the effects estimates are not constrained to have the same sign, which is not the case for the SAR model (Elhorst 2010).

Table 3 contains the direct, indirect (or spillover), and total effects from the SDM panel data model. We first note that all of the explanatory variables have been transformed by taking logs, which facilitates the interpretation of the effects estimates as elasticities. We also note that we utilized a five nearest-neighbor spatial weight matrix W in our empirical analysis as the average number of contiguous neighbors for U.S. states is five.[10]

The top portion of Table 3 contains the direct effects, which measure how much the dependent variable changes in a state when a particular explanatory variable changes in that same state. The first thing to note is that out of the seven included explanatory variables, four of them are significant at the 1% level (i.e., investment, black, Hispanic, metro), and income inequality and economic freedom are not statistically significant. Our main variable of interest is the economic freedom variable, which measures the level of economic freedom in each state. The results indicate that as the economic freedom index increases by 10%, GSP increases by 3.01%, which is fairly substantial, although the results are not statistically significant.

It is important to note that these are just the direct effects of an increase in economic freedom. While state policymakers care about the impact of changes in policy in their state, changes in the policies of other states also can influence their policies and economic outcomes. Since economic freedom spills over to nearby states (likely because of yardstick or Tiebout competition), it is important to account for how an increase in one state's economic freedom affects neighboring states' real per-capita GSP. As mentioned earlier, one of the features of the spatial Durbin panel model is its ability to empirically estimate the indirect or spillover effects of changes in explanatory variables. Essentially, these spillover effects measure how changes in our explanatory variables affect the dependent variable (real per-capita GSP) of surrounding states. The middle portion of Table 3 shows the indirect or spillover effects of changes in our explanatory variables. It is important to note that the scalar summary used to calculate the indirect effects summarizes spillovers over all states in the sample. Thus the spillover falling on any one state is much smaller than the total indirect effect. From a viewpoint within a state, these spillover effects might not be directly relevant. From a macro perspective, however, the fact that increases in economic freedom in Oregon directly increase economic growth in Oregon *and* indirectly in other states is important.

Of the seven included explanatory variables, five are statistically significant: economic freedom, percent black population, metro, and 90/10 income inequality are significant at the 1% level, while education is statistically significant at the 5% level. Again, our main variable of interest is the economic freedom index and the results indicate that as the economic freedom in a particular state increases by 10%, real per-capita GSP in *surrounding* states increases by 4.7%, on average. This is an important result for two reasons. First, this is the first study to use spatial panel data methods to estimate the relationship between economic freedom and GSP that correctly calculates the proper marginal effects of explanatory variables on growth. Second, we show that there are positive spillovers in economic freedom. A state increasing economic freedom, while not directly influencing their own real per-capita GSP, will have a ripple effect on all other states that will lead to higher incomes elsewhere. *Papers that utilize the state-level economic freedom index as an explanatory variable and do not control for spatial dependence are likely understating the true effect of economic freedom on the dependent variable.*

Finally, the bottom portion of Table 3 shows the point estimates for the total effects, which are defined as the sum of the direct and indirect effects. The total effects are all statistically significant at conventional levels, with the exception of the education variable, which is insignificant as the direct and indirect effects almost exactly offset each other. The total effect on the economic freedom variable has a point estimate of 0.504 which indicates that a 10% increase in the economic freedom index in the own and surrounding states increases GSP by 5.04%. Calculated at the mean real per-capita GSP, this means that a state like Oregon becoming as economically free as Tennessee would see $1,890 in additional GSP per-capita. Failure to account for the indirect effects of economic

10. For the purpose of our paper, the specification of the spatial weight matrix is generally not important (LeSage and Pace 2014).

TABLE 4
Spatial Durbin Panel Model Results with EFNA Components

	Coefficient	*t*-stat	*p* Value	
Direct				
Education	0.138	3.61	.00	***
Investment	1.211	41.71	.00	***
Size of government	−0.011	−0.63	.52	
Takings and discriminatory taxation	−0.005	−0.29	.76	
Regulation	0.043	2.07	.04	**
Black	0.035	4.28	.00	***
Hispanic	0.021	3.79	.00	***
Metro	−0.260	−5.02	.00	***
90/10 income inequality	0.012	1.40	.16	
Indirect				
Education	−0.130	−1.78	.08	*
Investment	0.027	0.44	.65	
Size of government	−0.003	−0.07	.94	
Takings and discriminatory taxation	0.291	6.31	.00	***
Regulation	0.082	1.77	.08	*
Black	−0.053	−2.95	.00	***
Hispanic	0.003	0.25	.79	
Metro	0.430	5.14	.00	***
90/10 income inequality	0.055	2.69	.00	***
Total				
Education	0.008	0.11	.91	
Investment	1.239	18.69	.00	***
Size of government	−0.014	−0.29	.76	
Takings and discriminatory taxation	0.285	5.30	.00	***
Regulation	0.125	2.41	.01	**
Black	−0.018	−0.96	.33	
Hispanic	0.025	1.77	.08	*
Metro	0.170	2.16	.03	**
90/10 income inequality	0.068	2.95	.00	***
Spatial autocorrelation δ	0.083	2.16	.03	**
R^2	.98			
Observations	1,536			

Notes: Dependent variable is real per-capita GSP. Size of Government, Takings and Discriminatory Taxation, and Regulation represent Areas 1, 2, and 3 of the *Economic Freedom of North America*, respectively. All variables are in log form.
* Significant at the 10% level, ** significant at the 5% level, *** significant at the 1% level.

freedom on growth, even if spatial dependence were controlled for, would underestimate the total effect of increases in economic freedom of real per-capita GSP by approximately $1,775.

In Table 4 we repeat the above analysis but break down economic freedom into its three component areas as defined by Stansel, Torra, and McMahon (2014): Size of Government, Takings and Discriminatory Taxation, and Regulation.[11] As an aggregate index, the components of the EFNA are likely to have heterogeneous effects and that is what we find. Decreases in the size of government (higher scores in Size of Government) have no statistically significant direct, indirect, or total effect on state income levels. This finding is consistent with a literature showing that government size, while a component of economic freedom, is different than the other areas of economic freedom because it can reflect both support and hindrance of other freedoms or productive activity (Beaulier et al. 2016).

Higher scores on Takings and Discriminatory Taxation (reflecting lower levels of takings and taxation) are not directly related to income levels in a statistically significant manner, but indirectly and in total the variable is positively related at the 1% level. The coefficient on Takings and Discriminatory Taxation is 0.285, suggesting that a 10% increase in that area's score is associated with a 2.85% increase in real per-capita state GSP, or just over $1,000 calculated at the mean. Lower regulation is positively and statistically significantly related to higher levels of real per-capita state GSP directly, indirectly, and in total. The total magnitude of the change

11. We are indebted to a referee for suggesting this addition. For further details on the areas and their components, we refer the reader to the extensive discussion in Stansel, Torra, and McMahon (2014).

is smaller than that of takings and discriminatory taxation, however. Looking at the bottom portion of Table 4, the coefficient on Regulation is 0.125. A 10% increase in a state's Regulation score (decline in regulation), is associated with a 1.25% increase in state income levels.[12] In addition to highlighting that not all areas of economic freedom are created equal, the results in the top portion of Table 4 help to explain the lack of direct effect of changes in economic freedom found in Table 3.

IV. CONCLUSION

The purpose of this paper is to contribute to the literature on the economic freedom-growth relationship and the literature using state-level economic freedom as an explanatory variable. We do so by detecting the presence and estimating the magnitude of spatial effects in a common empirical model of freedom and growth. A number of spatial specification tests were conducted and the prevailing specification is the SDM, which allows for spatial dependence in the dependent variable as well as allowing for the empirical estimation of spillover effects.

The empirical results from the spatial Durbin panel model are important for at least three reasons. First, we find evidence of a modest, although highly statistical significant, level of spatial autocorrelation, which indicates that there is a spatial pattern in real per-capita GSP. Second, we find empirical evidence that as the economic freedom in a state increases, there is a positive total effect on real per-capita GSP. Combined with the existing literature on the economic freedom-income relationship that is able to address causality, there should be little doubt about the positive association between economic freedom and income. From a policy perspective, however, our failure to find statistically significant direct effects of economic freedom is certainly important in understanding the incentives of policymakers, as the benefits of increases accrue elsewhere in the national economy. Further attempts to look closer at how economic freedom contributes to growth should utilize spatial econometric approaches where possible in order to properly obtain the marginal effects of individual components. In addition, when the Durbin model is the appropriate model, the indirect effects should highlight important spillovers across states related to individual policies that are components of economic freedom. For example, state and local jurisdictions clearly mimic each other in tax policy (Hall and Ross 2010).

Third, for the growing number of papers using the EFNA as an explanatory variable it is important to note the ability to identify and measure the spillover effects of changes in explanatory variables. One of the hallmarks of spatial econometric panel data models is that they can empirically estimate these indirect effects. These spillover effects measure how changes in an explanatory variable in one state affect the dependent variable in surrounding states. Our empirical results indicate that as economic freedom in one state increases, there is an increase in the real per-capita GSP in surrounding states. For papers in the broader economic freedom literature, this suggests that increases in economic freedom could indirectly effect entrepreneurship (Barnatchez and Lester 2017), income inequality (Apergis, Dincer, and Payne 2014), or any number of other variables. Economists and policymakers should appropriately estimate these spillovers and account for them in policy proposals.

12. It is important to note that the regulatory component of the EFNA is comprised entirely of labor-related regulations such as union density. We attempted to look at a broader measure of regulation provided in Ruger and Sorens (2016). Unfortunately, there were not enough observations available to run a comparable model to see if only regulations related to labor are what are important to income levels.

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